Recall convolution:

$$y[n] = h[n] * x[n] = \sum_{m=-\infty}^{\infty} h[m]\, x[n-m]$$

$$= \sum_{m=0}^{L-1} h[m]\, x[n-m]$$

For FIR filters, filter length is finite, and we assume filter length to be L

$\because$ $h[n]$ is causal $\therefore$ $n \geq 0$

however, we have n=-20:20;

What happens when $n < 0$?

$$y[n] = \sum_{m=-20}^{20} h[m]\, x[n-m]$$

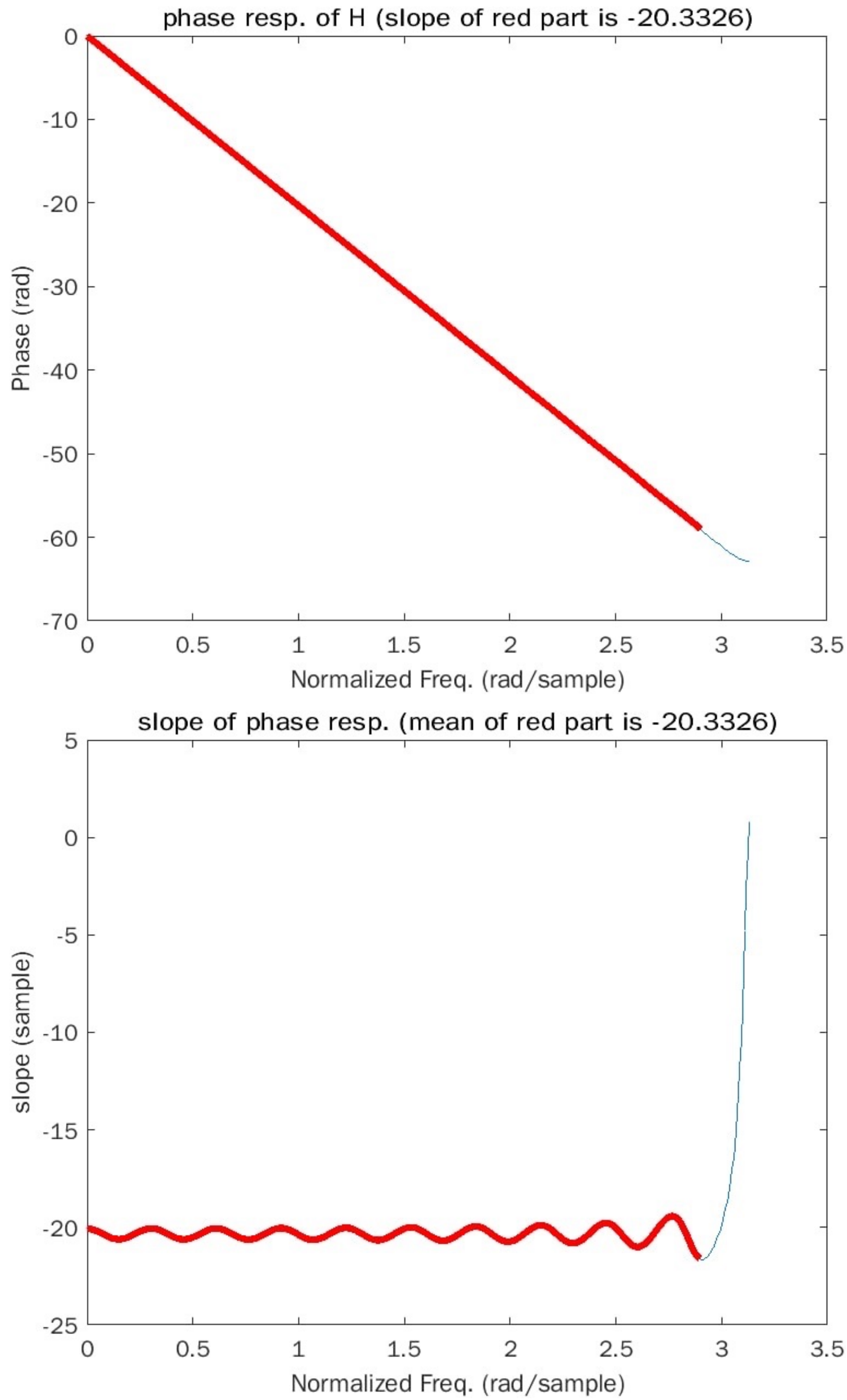
$$= h[-20]\, x[n+20] + h[-19]\, x[n+19] + \cdots$$

1. In this equation, current output is determined by future inputs, which is acausal in time-domain.
2. Now we create $g[n] = h[n-20]$, that is, we shift $h[n]$ 20 samples rightward, ~~to~~ make it causal.
3. From shift theorem, if $H(e^{j\omega}) = \sum_n h[n] e^{-j\omega n}$, then for $g[n] = h[n-20]$, $G(e^{j\omega}) = e^{-j\omega 20} H(e^{j\omega})$

$$= e^{-j\omega(20 + 1/3)}$$

$$\frac{\partial \measuredangle G}{\partial \omega} = -(20 + \tfrac{1}{3})$$

Also, MATLAB freqz interprets input to have an index starting from 0; therefore our $h[n]$ is shifted in advance!



phase resp. of H (slope of red part is -20.3326)
Phase (rad)
0
-10
-20
-30
-40
-50
-60
-70
0
0.5
1
1.5
2
2.5
3
3.5
Normalized Freq. (rad/sample)
slope of phase resp. (mean of red part is -20.3326)
slope (sample)
5
0
-5
-10
-15
-20
-25
0
0.5
1
1.5
2
2.5
3
3.5
Normalized Freq. (rad/sample)

MATLAB code

```
clc,clear,close all

% calculate h[n]
n = -20:20;
h = ( exp( 1i*pi*(n-1/3) ) - exp( -1i*pi*(n-1/3) ) ) ./...
    ( 2*pi*1i*(n-1/3) );

% plot phase response
figure,
[H,w] = freqz(h);
plot(w,phase(H))
hold on
num = 474;
tmp = phase(H);
plot(w(1:num),phase(H(1:num)),'r','LineWidth',3)
title(strcat("phase resp. of H (slope of red part is ",num2str( (tmp(num)-tmp(1)) / (w(num)-
w(1)) ),")"))
xlabel('Normalized Freq. (rad/sample)')
ylabel('Phase (rad)')
hold off


% plot the slope of phase response
figure,
plot(w(1:end-1),diff(phase(H))./diff(w))
xlabel('Normalized Freq. (rad/sample)')
ylabel('slope (sample)')
num = 38; % how many samples to be discarded
tmp = diff(phase(H))./diff(w);
hold on
plot(w(1:end-1-num),tmp(1:end-num),'r','LineWidth',3)
title(strcat("slope of phase resp. (mean of red part is ",...
      num2str(mean(tmp(1:end-num))),")"))
```